**University of Florida, Department of Mathematics**

# Analysis First-Year Exam, January 2026, Part 1

*Answer FOUR questions in detail. State carefully any results used without proof.*

**1.** Prove

(a) that every real sequence has a monotonic subsequence

(b) that every bounded real sequence has a convergent subsequence.

**2.** Let $T$ be a two-point metric space. Prove that the metric space $X$ is connected if and only if each continuous function from $X$ to $T$ is constant.

**3.** Let $X \subseteq \mathbb{R}$ and consider the following statements: Does (a) imply (b)? Does (b) imply (a)? Proof or counterexample for each.

(a) each continuous function from $X$ to $\mathbb{R}$ is bounded;

(b) $X$ is compact.

**4.** Let $f : \mathbb{R} \to \mathbb{R}$ be uniformly continuous and let $B \subseteq \mathbb{R}$ be bounded. Must $f(B)$ be bounded? Proof or counterexample.

**5.** The continuous function $f : \mathbb{R} \to \mathbb{R}$ is differentiable except perhaps at the point $a$. Decide whether $f$ is necessarily differentiable at $a$ if it is further assumed that $f'(t) \to 0$ as $t \to a$.